

Two sample tests

Two sample test when sample size is large

Summary of hypothesis test for $\mu_1 - \mu_2$ for large samples (n_1 and $n_2 \geq 30$)

We want to test:

$$H_0: \mu_1 - \mu_2 = D_0$$

versus

$$H_a: \begin{cases} \mu_1 - \mu_2 > D_0, & \text{upper tailed test} \\ \mu_1 - \mu_2 < D_0, & \text{lower tailed test} \\ \mu_1 - \mu_2 \neq D_0, & \text{two-tailed test.} \end{cases}$$

The TS is:

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - D_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

Two sample test when sample size is large

Replace σ_i with s_i , if σ_i , $i = 1, 2$, are not known. The RR is:

$$RR: \begin{cases} z > z_{\alpha}, & \text{upper tail RR} \\ z < -z_{\alpha}, & \text{lower tail RR} \\ |z| > z_{\alpha/2}, & \text{two tail RR,} \end{cases}$$

where z is the observed TS given by:

$$z = \frac{\bar{x}_1 - \bar{x}_2 - D_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

Assumption: The samples are independent and n_1 and $n_2 \geq 30$.

Decision: Reject H_0 , if the TS falls in the RR, and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, do not reject H_0 because there is not enough evidence to conclude that H_a is true for a given α and more data are needed.

Two sample test when sample size is large

EXAMPLE 6.5.1

In a salary equity study of faculty at a certain university, sample salaries of 50 male assistant professors and 50 female assistant professors yielded the following basic statistics.

	Sample mean salary	Sample standard deviation
Male assistant professor	\$46,400	360
Female assistant professor	\$46,000	220

Test the hypothesis that the mean salary of male assistant professors is more than the mean salary of female assistant professors at this university. Use $\alpha = 0.05$.

Solution

Let μ_1 be the true mean salary for male assistant professors and μ_2 be the true mean salary for female assistant professors at this university. To test:

$$H_0: \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a: \mu_1 - \mu_2 > 0$$

the TS is:

$$z = \frac{\bar{x}_1 - \bar{x}_2 - D_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{46,400 - 46,000}{\sqrt{\frac{(360)^2}{50} + \frac{(220)^2}{50}}} = 6.704.$$

The RR for $\alpha = 0.05$ is $\{z > 1.645\}$.

Because $z = 6.704 > 1.645$, we reject the null hypothesis at $\alpha = 0.05$. We conclude that the salary of male assistant professors at this university is higher than that of female assistant professors for $\alpha = 0.05$. Note that even though σ_1^2 and σ_2^2 are unknown, because $n_1 \geq 30$ and $n_2 \geq 30$, we could replace σ_1^2 and σ_2^2 with the respective sample variances. We are assuming that the salaries of male and female assistant professors are sampled independent of each other.

Two sample test when sample size is not large: Variance equal

Comparison of two population means, small sample case (pooled t -test)

We want to test:

$$H_0: \mu_1 - \mu_2 = D_0$$

versus

$$\begin{aligned} \mu_1 - \mu_2 &> D_0, && \text{upper tailed test} \\ H_a: \mu_1 - \mu_2 &< D_0, && \text{lower tailed test} \\ \mu_1 - \mu_2 &\neq D_0, && \text{two-tailed test.} \end{aligned}$$

The TS is:

$$T = \frac{\bar{X}_1 - \bar{X}_2 - D_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Here the pooled sample variance is:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Then the RR is:

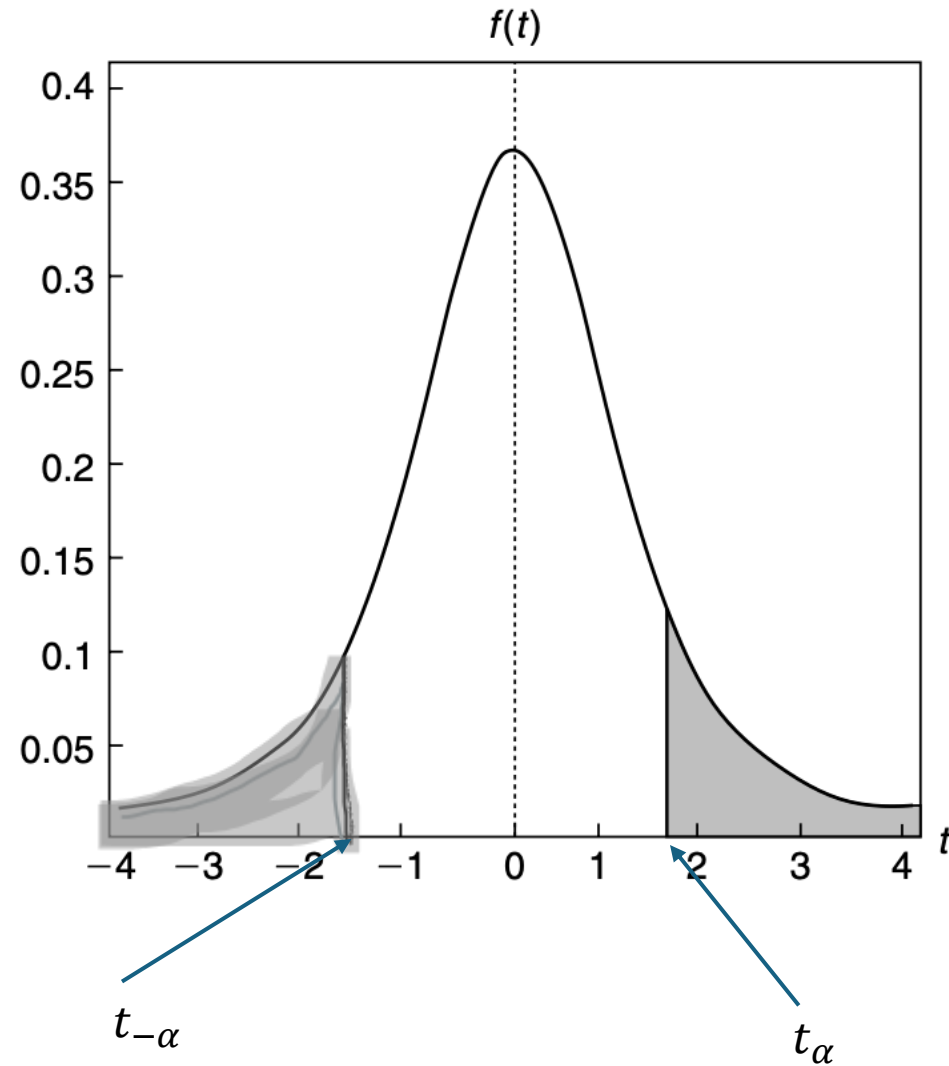
$$RR: \begin{cases} t > t_\alpha, & \text{upper tailed test} \\ t < -t_\alpha, & \text{lower tail test} \\ |t| > t_{\alpha/2}, & \text{two-tailed test} \end{cases}$$

where t is the observed TS and t_α is based on $(n_1 + n_2 - 2)$ degrees of freedom, and such that $P(T > t_\alpha) = \alpha$.

Decision: Reject H_0 , if TS falls in the RR, and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, do not reject H_0 because there is not enough evidence to conclude that H_a is true for a given α .

Assumptions: The samples are independent and come from normal populations with means μ_1 and μ_2 , and with (unknown) equal variances, that is, $\sigma_1^2 = \sigma_2^2$.

Two sample test when sample size is not large: Variance equal



Two sample test when sample size is not large: Variance equal

EXAMPLE 6.5.2

The IQs of 17 students from one area of a city showed a sample mean of 106 with a sample standard deviation of 10, whereas the IQs of 14 students from another area chosen independently showed a sample mean of 109 with a sample standard deviation of 7. Is there a significant difference between the IQs of the two groups at $\alpha = 0.01$? Assume that the population variances are equal.

Solution

We need to test:

$$H_0: \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a: \mu_1 - \mu_2 \neq 0.$$

Here $n_1 = 17$, $\bar{x}_1 = 106$, and $s_1 = 10$. Also, $n_2 = 14$, $\bar{x}_2 = 109$, and $s_2 = 7$.

We have:

$$\begin{aligned} s_p^2 &= \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \\ &= \frac{(16)(10)^2 + (13)(7)^2}{29} = 77.138. \end{aligned}$$

The TS is:

$$t = \frac{\bar{x}_1 - \bar{x}_2 - D_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{106 - 109}{(\sqrt{77.138}) \sqrt{\frac{1}{17} + \frac{1}{14}}} = -0.94644.$$

For $\alpha = 0.01$, $t_{0.01,29} = 2.462$. Hence, the RR is $t < -2.462$ or $t > 2.462$.

Because the observed value of the TS, $t = -0.94644$, does not fall in the RR, there is not enough evidence to conclude that the mean IQs are different for the two groups. Here we assume that the two samples are independent and taken from normal populations.

Two sample test when sample size is not large: Variance unequal

$$H_0: \mu_1 - \mu_2 = D_0 \text{ vs. } H_a: \begin{cases} \mu_1 - \mu_2 > D_0 \\ \mu_1 - \mu_2 < D_0 \\ \mu_1 - \mu_2 \neq D_0 \end{cases}$$

$$T_v = \frac{\bar{X}_1 - \bar{X}_2 - D_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$v = \frac{\left[\frac{(s_1^2/n_1) + (s_2^2/n_2)}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \right]^2}{\frac{s_1^4}{n_1^2(n_1 - 1)} + \frac{s_2^4}{n_2^2(n_2 - 1)}}$$

▮ nearest integer less than
 v should be considered

Welch's t-test or Smithe Satterthwaite procedure

Two sample test when sample size is not large: Variance unequal

EXAMPLE 6.5.3

Assume that two populations are normally distributed with unknown and unequal variances. Two independent samples were drawn from these populations and the data obtained resulted in the following basic statistics:

$$\begin{aligned}n_1 &= 18 & \bar{x}_1 &= 20.17 & s_1 &= 4.3 \\n_2 &= 12 & \bar{x}_2 &= 19.23 & s_2 &= 3.8.\end{aligned}$$

Test at the 5% level of significance whether the two population means are different.

Solution

We need to test the hypothesis:

$$H_0: \mu_1 - \mu_2 = 0 \text{ versus } H_a: \mu_1 - \mu_2 \neq 0.$$

Here $n_1 = 18$, $\bar{x}_1 = 20.17$, and $s_1 = 4.3$. Also, $n_2 = 12$, $\bar{x}_2 = 19.23$, and $s_2 = 3.8$.

The degrees of freedom for the t distribution are given by:

$$\begin{aligned}v &= \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \\&= \frac{\left(\frac{(4.3)^2}{18} + \frac{(3.8)^2}{12}\right)^2}{\frac{\left(\frac{(4.3)^2}{18}\right)^2}{17} + \frac{\left(\frac{(3.8)^2}{12}\right)^2}{11}} = 25.685.\end{aligned}$$

Two sample test when sample size is not large: Variance unequal

Hence, rounding down we have $v = 25$ degrees of freedom. For $\alpha = 0.05$, $t_{0.025,25} = 2.060$. Thus, the RR is $t < -2.060$ or $t > 2.060$.

The TS is given by:

$$\begin{aligned} t_v &= \frac{\bar{x}_1 - \bar{x}_2 - D_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \\ &= \frac{20.17 - 19.23}{\sqrt{\frac{(4.3)^2}{18} + \frac{(3.8)^2}{12}}} = 0.62939. \end{aligned}$$

Because the observed value of the TS, $t_v = 0.62939$, does not fall in the RR, we do not reject the null hypothesis. At $\alpha = 0.05$ there is not enough evidence to conclude that the population means are different. Note that the assumptions we made are that the samples are independent and came from two normal populations. No homogeneity assumption of the variance is made.

Two sample test : Testing equality of proportion

Summary of hypothesis test for $(p_1 - p_2)$ for large samples ($n_i p_i > 5$ and $n_i q_i > 5$, for $i = 1, 2$)

To test:

$$H_0: p_1 - p_2 = D_0$$

versus

$$p_1 - p_2 < D_0, \quad \text{upper tailed test}$$

$$H_a: p_1 - p_2 > D_0, \quad \text{lower tailed test}$$

$$p_1 - p_2 \neq D_0, \quad \text{two-tailed test}$$

at the level of significance α , the TS is:

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - D_0}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}}$$

where z is the observed value of Z .

The RR is:

$$RR: \begin{cases} z > z_\alpha, & \text{upper tailed RR} \\ z < -z_\alpha, & \text{lower tailed RR} \\ |z| > z_{\alpha/2}, & \text{two-tailed RR} \end{cases}$$

Assumption: The samples are independent and

$$n_i p_i > 5 \text{ and } n_i q_i > 5, \text{ for } i = 1, 2.$$

Decision: Reject H_0 if the TS falls in the RR, and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, do not reject H_0 , because there is not enough evidence to conclude that H_a is true for a given α and more data are needed.

Two sample test : Testing equality of proportion

EXAMPLE 6.5.5

Because of the impact of the global economy on a high-wage country such as the United States, it is claimed that the domestic content in manufacturing industries fell between 1977 and 1997. A survey of 36 randomly picked US companies gave the proportion of domestic content total manufacturing in 1977 as 0.37 and in 1997 as 0.36. At the 1% level of significance, test the claim that the domestic content really fell during the period 1977–97.

Solution

Let p_1 be the domestic content in 1977 and p_2 be the domestic content in 1997.

Given $n_1 = n_2 = 36$, $\hat{p}_1 = 0.37$ and $\hat{p}_2 = 0.36$. We need to test:

$$H_0: p_1 - p_2 = 0 \quad \text{vs.} \quad H_a: p_1 - p_2 > 0.$$

The TS is:

$$\begin{aligned} z &= \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}} \\ &= \frac{0.37 - 0.36}{\sqrt{\frac{(0.37)(0.63)}{36} + \frac{(0.36)(0.64)}{36}}} = 0.08813. \end{aligned}$$

For $\alpha = 0.01$, $z_{0.01} = 2.325$. Hence, the RR is $z > 2.325$.

Because the observed value of the TS does not fall in the RR, at $\alpha = 0.01$, there is not enough evidence to conclude that the domestic content in manufacturing industries fell between 1977 and 1997.

Two sample test : Testing equality of variance

Testing for the equality of variances

To test:

$$H_0: \sigma_1^2 = \sigma_2^2$$

versus

$$\sigma_1^2 > \sigma_2^2, \quad \text{lower tailed test}$$

$$H_a: \sigma_1^2 < \sigma_2^2, \quad \text{upper tailed test}$$

$$\sigma_1^2 \neq \sigma_2^2, \quad \text{two-tailed test}$$

at significance level α , the TS is:

$$F = \frac{S_1^2}{S_2^2}.$$

The RR is:

$$RR: \begin{cases} f > F_{\alpha}(v_1, v_2), & \text{upper tailed RR} \\ f < F_{1-\alpha}(v_1, v_2), & \text{lower tailed RR} \\ f > F_{\alpha/2}(v_1, v_2) \text{ or } f < F_{1-\alpha/2}(v_1, v_2), & \text{two-tailed RR} \end{cases}$$

where f is the observed TS given by $f = \frac{s_1^2}{s_2^2}$.

Decision: Reject H_0 if the TS falls in the RR and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, keep H_0 , because there is not enough evidence to conclude that H_a is true for a given α and more data are needed.

Assumptions:

- (i) The two random samples are independent.
- (ii) Both populations are normal.

$$\text{Recall : } F_{1-\alpha}(v_1, v_2) = \frac{1}{F_{\alpha}(v_1, v_2)}$$

Two sample test : Testing equality of variance

EXAMPLE 6.5.6

Consider two independent random samples, $X_1, \dots, X_n$ from an $N(\mu_1, \sigma_1^2)$ distribution and $Y_1, \dots, Y_n$ from an $N(\mu_2, \sigma_2^2)$ distribution. Test $H_0 : \sigma_1^2 = \sigma_2^2$ versus $H_a : \sigma_1^2 \neq \sigma_2^2$ for the following basic statistics:

$$n_1 = 25, \bar{x}_1 = 410, s_1^2 = 95, \quad \text{and} \quad n_2 = 16, \bar{x}_2 = 390, s_2^2 = 300.$$

Use $\alpha = 0.20$.

Solution

Test $H_0 : \sigma_1^2 = \sigma_2^2$ versus $H_a : \sigma_1^2 \neq \sigma_2^2$. This is a two-tailed test.

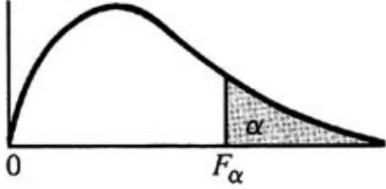
Here the degrees of freedom are $\nu_1 = 24$ and $\nu_2 = 15$. The TS is:

$$F = \frac{s_1^2}{s_2^2} = \frac{95}{300} = 0.317.$$

From the F table with $\alpha/2 = 0.10$, $F_{0.10}(24, 15) = 1.90$ and $F_{0.90}(24, 15) = (1/F_{0.10}(15, 24)) = 1/1.78 = 0.56$.

Hence, the RR is $F > 1.90$ or $F < 0.56$. Because the observed value of the TS, 0.317, is less than 0.56, we reject the null hypothesis. There is evidence that the population variances are not equal.

TABLE AV.5 Percentage point of F -distributions.



Denominator d.f.	Numerator d.f.									
	α	1	2	3	4	5	6	7	8	9
1	.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86
	.050	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5
	.025	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3
	.010	4052	4999.5	5403	5625	5764	5859	5928	5982	6022
	.005	16211	20000	21615	22500	23056	23437	23715	23925	24091
2	.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38
	.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
	.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39
	.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
	.005	198.5	199.0	199.2	199.2	199.3	199.3	199.4	199.4	199.4
3	.100	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24
	.050	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
	.025	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47
	.010	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35
	.005	55.55	49.80	47.47	46.19	45.39	44.84	44.43	44.13	43.88
4	.100	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94
	.050	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
	.025	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90
	.010	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
	.005	31.33	26.28	24.26	23.15	22.46	21.97	21.62	21.35	21.14

TABLE AV.5 Percentage point of F -distributions.—cont'd

F_{α}											d.f.
Numerator d.f.											
10	12	15	20	24	30	40	60	120	∞	α	
2.06	2.02	1.97	1.92	1.90	1.87	1.85	1.82	1.79	1.76	.100	15
2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07	.050	
3.06	2.96	2.86	2.76	2.70	2.64	2.59	2.52	2.46	2.40	.025	
3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87	.010	
4.42	4.25	4.07	3.88	3.79	3.69	3.58	3.48	3.37	3.26	.005	
2.03	1.99	1.94	1.89	1.87	1.84	1.81	1.78	1.75	1.72	.100	16
2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01	.050	
2.99	2.89	2.79	2.68	2.63	2.57	2.51	2.45	2.38	2.32	.025	
3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75	.010	
4.27	4.10	3.92	3.73	3.64	3.54	3.44	3.33	3.22	3.11	.005	
2.00	1.96	1.91	1.86	1.84	1.81	1.78	1.75	1.72	1.69	.100	17
2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96	.050	
2.92	2.82	2.72	2.62	2.56	2.50	2.44	2.38	2.32	2.25	.025	
3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65	.010	
4.14	3.97	3.79	3.61	3.51	3.41	3.31	3.21	3.10	2.98	.005	
1.98	1.93	1.89	1.84	1.81	1.78	1.75	1.72	1.69	1.66	.100	18
2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92	.050	
2.87	2.77	2.67	2.56	2.50	2.44	2.38	2.32	2.26	2.19	.025	
3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57	.010	
4.03	3.86	3.68	3.50	3.40	3.30	3.20	3.10	2.99	2.87	.005	
1.96	1.91	1.86	1.81	1.79	1.76	1.73	1.70	1.67	1.63	.100	19
2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88	.050	
2.82	2.72	2.62	2.51	2.45	2.39	2.33	2.27	2.20	2.13	.025	
3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49	.010	
3.93	3.76	3.59	3.40	3.31	3.21	3.11	3.00	2.89	2.78	.005	